Python: exercises on branching

This notebook was generated in solutions mode.

Exercise: Positive, Negative, or Zero

Exercise Description

Write a program that determines whether a given number is positive, negative, or zero.

Use the number `num = -5` and store the result in variable `result` ("positive", "negative", or "zero").

Your solution

```
num = -5
if num > 0:
    result = "positive"
elif num < 0:
    result = "negative"
```

```python
else:
    result = "zero"
```

Test your solution

```python
assert result == "negative"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Even or Odd Number

Exercise Description

Write a program that determines whether a given integer is even or odd.

Use the number `num = 7` and store the result in variable `result` ("even" or "odd").

Your solution

```
num = 7
if num % 2 == 0:
    result = "even"
else:
    result = "odd"
```

Test your solution

```
assert result == "odd"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Grade Evaluator

Exercise Description

Write a program that converts a numerical score (0-100) to a letter grade:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D
- 0-59: F

Use the score `score = 85` and store the letter grade in variable `result`.

Test your solution

```
assert result == "B"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Divisibility Checker

Exercise Description

Write a program that checks if a given number is divisible by both 3 and 5.

Use the number `num = 30` and store the result in variable `result` ("divisible by both" or "not divisible by both").

Your solution

```
num = 30
if num % 3 == 0 and num % 5 == 0:
    result = "divisible by both"
else:
    result = "not divisible by both"
```

Test your solution

```
assert result == "divisible by both"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Leap Year Checker

Exercise Description

Write a program that determines whether a given year is a leap year or not.

A leap year is:

- Divisible by 4 but not by 100, OR
- Divisible by 400

Use the year `year = 2024` and store the result in variable `result` ("leap year" or "not leap year").

Your solution

```python
year = 2024
if (year % 4 == 0 and year % 100 != 0) or year % 400 == 0:
    result = "leap year"
else:
    result = "not leap year"
```

Test your solution

```python
assert result == "leap year"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Vowel or Consonant

Exercise Description

Write a program that determines whether a given letter is a vowel or consonant.

Use the letter `letter = "A"` and store the result in variable `result` ("vowel" or "consonant").

Your solution

```
letter = "A"
if letter in "aeiouAEIOU":
    result = "vowel"
else:
    result = "consonant"
```

Test your solution

```
assert result == "vowel"
```

Debug your solution

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```
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```

Exercise: Password Validator

Exercise Description

Write a program that validates a password. If the password is "password123", grant access, otherwise deny access.

Use the password `password = "password123"` and store the result in variable `result` ("Access granted" or "Access denied").

Your solution

```
password = "password123"
if password == "password123":
```

```
    result = "Access granted"
else:
    result = "Access denied"
```

Test your solution

```
assert result == "Access granted"
```

Debug your solution

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```
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```

Exercise: Largest of Three Numbers

Exercise Description

Write a program that finds the largest of three given numbers.

Use the numbers `num1 = 15`, `num2 = 8`, and `num3 = 22`. Store the largest number in variable `result`.

Your solution

```
num1 = 15
num2 = 8
num3 = 22
if num1 >= num2 and num1 >= num3:
    result = num1
elif num2 >= num1 and num2 >= num3:
    result = num2
else:
    result = num3
```

Test your solution

```
assert result == 22
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Day of the Week

Exercise Description

Write a program that converts a number (1-7) to the corresponding day of the week:

- 1: Monday
- 2: Tuesday
- 3: Wednesday
- 4: Thursday
- 5: Friday
- 6: Saturday
- 7: Sunday
- Any other number: "Invalid number"

Use the number `day = 5` and store the day name in variable `result`.

Your solution

```
day = 5
if day == 1:
    result = "Monday"
elif day == 2:
```

```
    result = "Tuesday"
elif day == 3:
    result = "Wednesday"
elif day == 4:
    result = "Thursday"
elif day == 5:
    result = "Friday"
elif day == 6:
    result = "Saturday"
elif day == 7:
    result = "Sunday"
else:
    result = "Invalid number"
```

Test your solution

```
assert result == "Friday"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Simple Calculator

Exercise Description

Write a program that performs basic arithmetic operations (+, -, *, /) on two numbers based on the given operation.

Use `num1 = 10.0`, `num2 = 3.0`, and `operation = "+"`. Store the result in variable `result`. Handle division by zero with an error message.

Your solution

```
num1 = 10.0
num2 = 3.0
operation = "+"
if operation == "+":
    result = num1 + num2
elif operation == "-":
    result = num1 - num2
elif operation == "*":
    result = num1 * num2
elif operation == "/":
    if num2 != 0:
        result = num1 / num2
    else:
        result = "Error: Cannot divide by zero"
```

```
else:
    result = "Invalid operation"
```

Test your solution

```
assert result == 13.0
```

Debug your solution

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```
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```

Exercise: Triangle Type Classifier

Exercise Description

Write a program that determines the type of triangle based on the lengths of its three sides:

- Equilateral: All sides are equal
- Isosceles: Two sides are equal
- Scalene: No sides are equal

Use `side1 = 5.0`, `side2 = 5.0`, and `side3 = 8.0`. Store the triangle type in variable `result`.

Your solution

```
side1 = 5.0
side2 = 5.0
side3 = 8.0
if side1 == side2 == side3:
    result = "equilateral"
elif side1 == side2 or side2 == side3 or side1 == side3:
    result = "isosceles"
else:
    result = "scalene"
```

Test your solution

```
assert result == "isosceles"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Maximum of Three Numbers

Exercise Description

Write a program that finds the maximum (largest) of three given numbers.

Use the numbers `num1 = 12.5`, `num2 = 8.3`, and `num3 = 15.7`. Store the maximum number in variable `result`.

Your solution

```
num1 = 12.5
num2 = 8.3
num3 = 15.7
if num1 >= num2 and num1 >= num3:
    result = num1
elif num2 >= num1 and num2 >= num3:
    result = num2
else:
    result = num3
```

Test your solution

```
assert result == 15.7
```

Debug your solution